

SET → combine results of multiple queries.

Example

Query 1:

```
SELECT  
  <col-1>  
  <col-2>  
FROM Table-A
```

UNION

Query 2:

```
SELECT  
  <col-1>,  
  <col-2>  
FROM Table-B
```

SET Operator

→ Clauses that can go to
individual query

WHERE | JOIN | GROUP BY | HAVING

Rule 1

Clauses that go at the very end
of Set operated queries

ORDER BY

Rule 2

Number of columns must be equal.

Rule 3

Data type of columns must match or be compatible

Rule 4

Order of column in each query must be the same.

Rule 5

Name of the result column is determined by the first query.

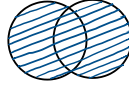
Rule 6

Column selection is manual. The user must match the appropriate columns to avoid incorrect results.

the appropriate columns to avoid incorrect results.

UNION

- Return all distinct rows
- No duplicates in the result

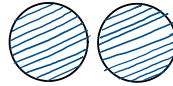


UNION ALL

- Return all rows
- Result can have duplicates hmm

UNION ALL is much faster than UNION

- No duplicate filtering needed.



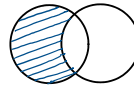
USE UNION ALL instead of UNION if UNION can be avoided. For instance, if one is certain that there are no duplicates.

EXCEPT

- All distinct rows in the first query not found in the second query

Remove results from the first query that also exist in the second query

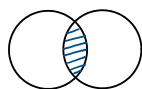
- Therefore order of the query matters.



Back

INTERSECT

- Return rows that are common in both queries.



USE Cases - UNION

Combine Information

Combine similar information before performing data analysis

Instead of performing many queries,
Combine data, then perform a single query

Working with Fragmented Data Tables

One data table can be fragmented;

- To optimize performance
- To archive older data.

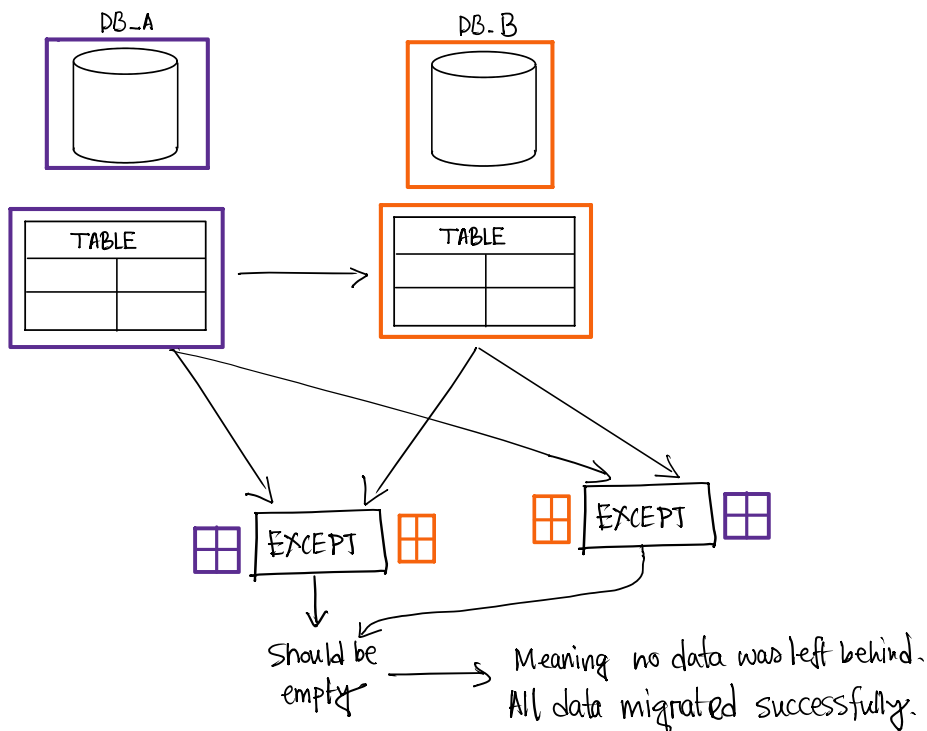
Use UNION to combine the sub-tables into one big table. Then query that instead of sub-tables.

USE Cases - EXCEPT

Delta Detection

Data Completeness Check

Check data completeness during data migration



SQL Developer Best Practices

- In final version of the code,

avoid using (*) to address all columns.

Explicitly write down column names.

Because, two different tables might not contain

similar columns (out of order, unmatch number of rows,

etc.) Check that, fix that before doing anything.